

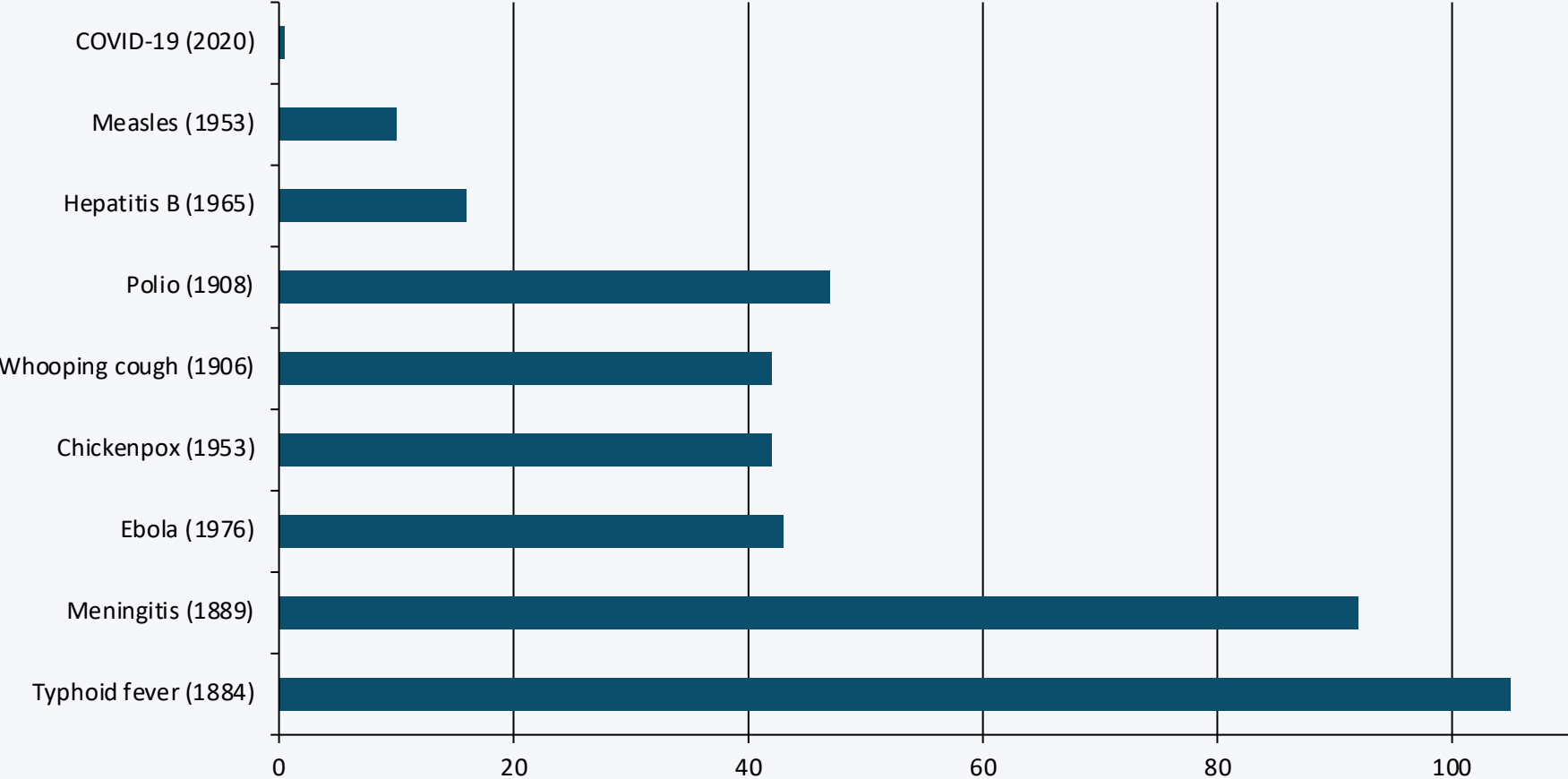
The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of two centuries of vaccine innovation, disease elimination, and remaining gaps

Source: Our World in Data | Roush & Murphy (2007)

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure, selected diseases (1884–2020)



Key Insight

Early vaccines took decades because scientists didn't yet understand germ theory.

Modern tools — genome sequencing, mRNA, virus-like particles — collapsed timelines to months.

COVID-19:
<1 year
(mRNA & genomic tools)

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

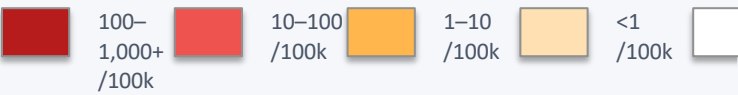
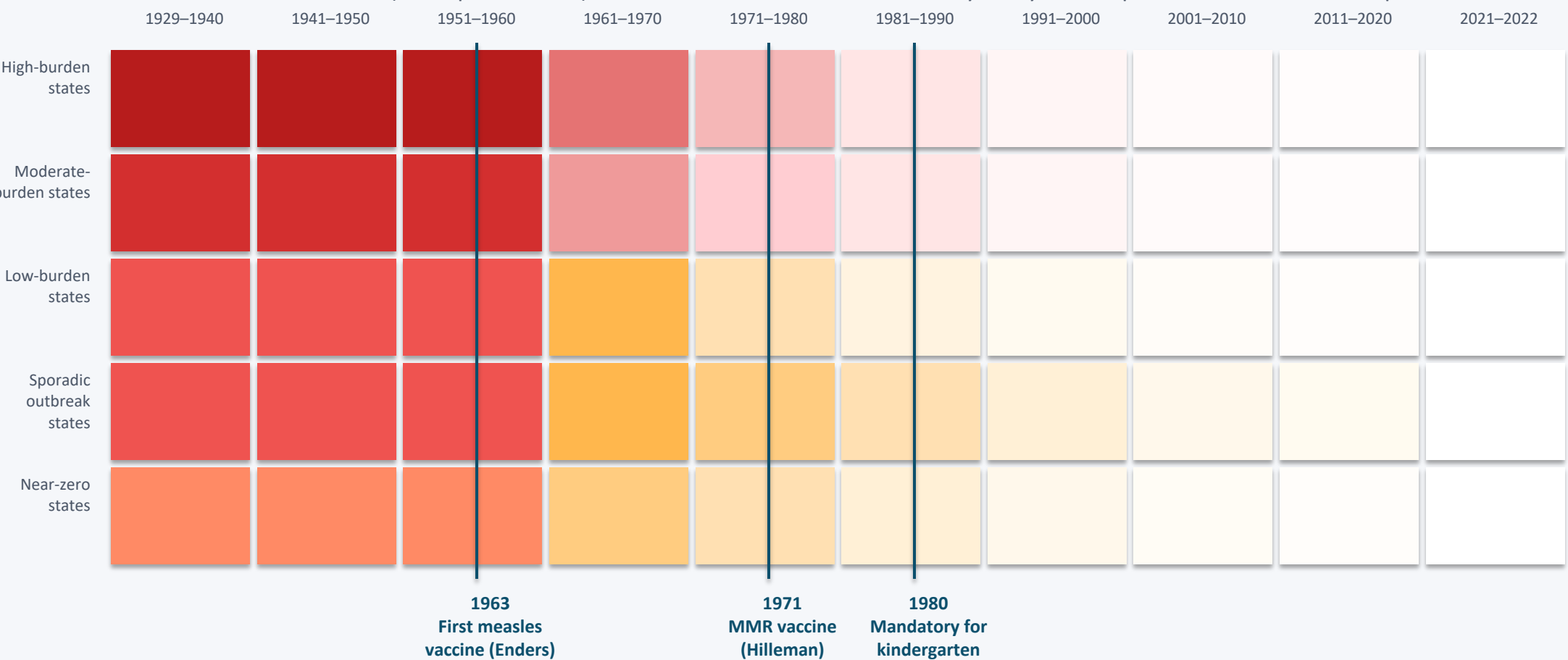
Pre-vaccine vs. post-vaccine annual cases and deaths per million population (Roush & Murphy, 2007)

Disease	Pre-Vaccine Cases /M/yr	Post-Vaccine Cases /M/yr	Case Reduction	Pre-Vaccine Deaths /M/yr	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Measles	3,044	0.2	99.99%	2.5	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Green = 100% elimination | Orange = 50–89% reduction | Red = <50% reduction

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

State-level measles incidence (cases per 100,000) from 1929–2022. Before 1963, nearly every state reported 100–1,000+ cases per 100,000.



Smallpox: The Only Human Disease Eradicated Through Vaccination



Why smallpox was uniquely suited for eradication

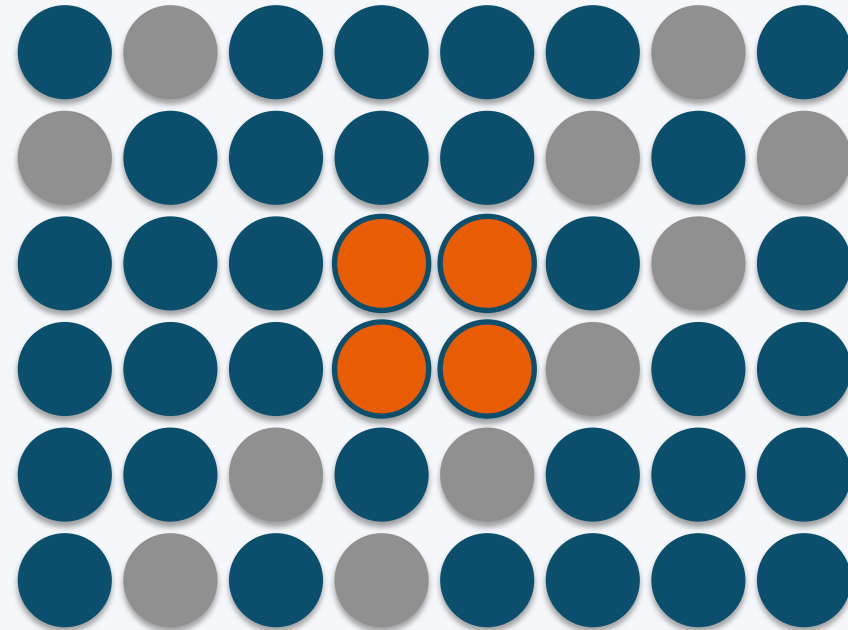
- Virus infected only humans — no animal reservoir
- Symptoms were easily distinguishable from other diseases
- Vaccine was cheap, stable, and highly effective
- Global political will via WHO resolution and funding

100%
elimination of cases and deaths

Smallpox remains the only human disease fully eliminated through vaccination — certified in 1980 after a 21-year intensified campaign.

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

When enough people are vaccinated, pathogens cannot spread effectively — shielding those who cannot be vaccinated.



Vaccinated



Unvaccinated
(susceptible)



Vulnerable (cannot be
vaccinated)

The Shield Effect

Newborns too young for certain vaccines

Immunocompromised individuals (HIV, transplant recipients)

Cancer patients on chemotherapy

People with severe allergies to vaccine components

High vaccination coverage creates a barrier that breaks chains of transmission, indirectly protecting the most vulnerable members of society.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Not all vaccines have achieved near-complete control. Coverage, access, and trust remain critical barriers.

Pneumococcal Disease

40.5% case reduction
31.3% death reduction

Lowest impact among major vaccines.
Improved conjugate vaccines and broader pediatric coverage needed.

Hepatitis A & B

89% case reduction (A)
83.9% case reduction (B)

Incomplete control despite effective vaccines. Access gaps in underserved populations and inconsistent adult booster uptake.

Global Access & Trust

Supply shortages
Coverage inequity

Not everyone has access to vaccines. Supply shortages put children at risk, and poor scientific understanding can reduce trust and uptake.

Priority actions: Expand pneumococcal conjugate coverage • Strengthen hepatitis A/B adult programs • Address supply-chain gaps • Invest in community trust-building

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

100% Elimination Achieved

Diphtheria, smallpox, and acute polio eliminated in the US. Measles and rubella reduced to near-zero with sporadic outbreaks only.

Development Timelines Collapsed

From 105 years (typhoid, 1884→1989) to under 1 year (COVID-19, 2020). mRNA and genomic tools are rewriting the speed of response.

Herd Immunity Protects Communities

Vaccination shields newborns, immunocompromised patients, and those on chemotherapy by breaking chains of transmission.

Coverage Gaps Persist

Pneumococcal disease (40.5% reduction), hepatitis A (89%), and hepatitis B (83.9%) still show incomplete control. Access, supply, and trust are barriers.

Call to Action for Global Health Systems

Prioritize equitable access • Modernize supply chains • Invest in next-generation vaccines • Build public trust through transparent science communication